Contact

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(LinkedIn)

Top Skills

Computational Biology
Virology
Functional Genomics

Languages

Assamese (Limited Working)
English (Native or Bilingual)
Hindi (Professional Working)

Honors-Awards

Gold Medal for Academic Excellence
Certificate of Merit of Outstanding Performance
Regional Topper - 11th National Cyber Olympiad
Jane Hill Prize in Computer Science
The Richard Howland Memorial Scholarship

Publications

Repeat expansions confer WRN dependence in microsatellite-unstable cancers
The role of EGR1 and AP1 in acute myeloid leukemia cell reprogramming toward cell cycle arrest and apoptosis
Transcriptional subtype-specific microenvironmental crosstalk and tumor cell plasticity in metastatic pancreatic cancer

Ashir Borah

PhD Candidate in Bioinformatics | Functional Genomics, Virology, CRISPR Screens, and AI-Enabled Research Systems
San Francisco, California, United States

Summary

I am a PhD candidate at UCSF and the Arc Institute working across functional genomics, virology, and computational biology. My research combines experimental and computational approaches to build and apply tools for studying gene regulation, host-virus interactions, and disease biology.My work spans CRISPR-based screening, assay development, high-dimensional data analysis, and platform building for biological discovery. In parallel, I build AI-enabled research systems, including LLM-based councils, research agents, APIs, and multi-agent workflows that support literature synthesis, hypothesis generation, experimental planning, and scientific reasoning.Previously, I worked at the Broad Institute on cancer target discovery using genome-scale perturbation data, statistical modeling, and machine learning. I am especially interested in problems where strong experimental design, thoughtful computation, and well-built tooling can reveal biology that is otherwise difficult to measure.

Experience

Arc Institute
PhD Researcher
June 2023 - Present (2 years 11 months)
Palo Alto, CA

Design and develop experimental and computational methods for functional genomics and virology

Build CRISPR-based screening and perturbation systems to study gene regulation, host-virus biology, and disease mechanisms

Develop assay platforms and analysis workflows for high-dimensional genomic datasets

Build AI-enabled scientific tooling, including LLM-based councils, APIs, and multi-agent workflows for literature synthesis, hypothesis generation, and experimental planning

Work at the interface of wet lab biology, computational analysis, and research infrastructure development

University of California, San Francisco

PhD Candidate, Biological and Medical Informatics

September 2022 - Present (3 years 8 months)

San Francisco, CA

Conduct PhD research in functional genomics, virology, and computational biology in collaboration with the Arc Institute

Develop computational and experimental approaches for studying gene regulation, perturbation biology, and disease mechanisms

Analyze large-scale biological datasets using statistical, machine learning, and high-dimensional data analysis methods

Led an R bootcamp for postdocs and clinicians, teaching foundational coding and biomedical data analysis to a cohort of 100+ trainees

Broad Institute of MIT and Harvard

2 years 10 months

Computational Associate II

July 2021 - April 2022 (10 months)

Cambridge, MA

Validated candidate cancer targets identified through genome-scale CRISPR knockout screens

Integrated computational analyses with experimental follow-up to support cancer target discovery

Developed approaches for prioritizing candidate therapeutic targets in cancer

Computational Associate I

July 2019 - July 2021 (2 years 1 month)

Cambridge, Massachusetts

Applied machine learning and statistical methods to identify vulnerabilities in cancer cell lines, with a focus on GI cancers

Analyzed perturbation and functional genomics datasets to support translational target discovery

Prioritized candidate cancer dependencies for downstream therapeutic investigation

Dickinson College
Teaching, Mentorship, and Research Roles
August 2015 - May 2019 (3 years 10 months)
Carlisle, Pennsylvania

Served as a teaching assistant for computer science courses including Introduction to Java and Data Structures

Tutored students in mathematics and computer science and developed individualized study support

Held mentorship and residential leadership roles focused on student support, advising, and community building

Completed a student research project on secure CoAP-DTLS systems for resource-constrained devices

Uliza
Machine Learning Developer
September 2017 - December 2017 (4 months)

Built machine learning-driven systems to support crowdsourced workflow automation

Helped set up core technical infrastructure, including server management, database optimization, and disaster recovery

Education

Dickinson College
Bachelor of Science (B.Sc.), Mathematics and Computer Science · (2015 - 2019)